Case Study 1: Restaurant Chain

Step 1 - Business process

- **Sales & revenue** (order lines), **costs** (operating costs per branch), **HR head-count** (employees per branch). Products variety is derived from the sales data.
- Goal: support expansion decisions by comparing branches, governorates and regions.

Step 2 - Grain

Fact table	Grain (one row =)	Type
Fact_Sales	One product line on one order (order item)	Transaction
Fact_Costs	One cost type, for one branch, in one month (Date_Key = first day of that month)	Periodic
Fact_Headcount	One branch, at the end of one month	Periodic snapshot

Step 3 - Dimensions

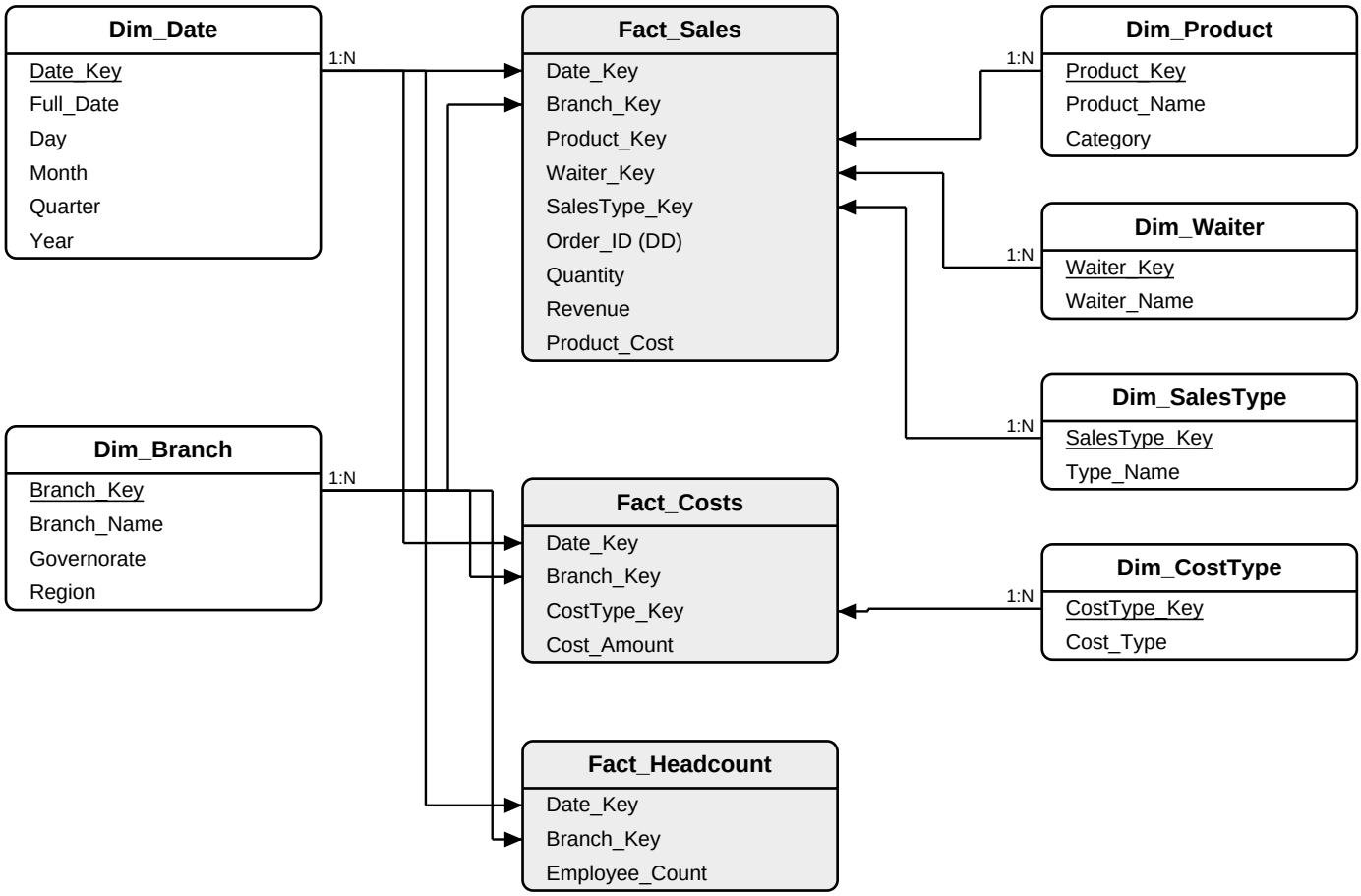
Dimension	Attributes	Hierarchy / use
Dim_Date	Date_Key, Full_Date, Day, Month, Quarter, Year	Day > Month > Quarter > Year
Dim_Branch	Branch_Key, Branch_Name, Governorate, Region	Branch > Governorate > Region
Dim_Product	Product_Key, Product_Name, Category	Product > Category (pizzas, sandwiches, main dishes, drinks...)
Dim_Waiter	Waiter_Key, Waiter_Name	Sales per waiter
Dim_SalesType	SalesType_Key, Type_Name	Delivery / Dine-in / Take-away
Dim_CostType	CostType_Key, Cost_Type	Rent, salaries, raw materials, utilities...

Step 4 - Facts (measures)

Fact table	Foreign keys	Measures
Fact_Sales	Date, Branch, Product, Waiter, SalesType	Quantity, Revenue (additive), Product_Cost, Order_ID (degenerate dimension)
Fact_Costs	Date, Branch, CostType	Cost_Amount (additive)
Fact_Headcount	Date, Branch	Employee_Count (semi-additive: not summed over time)

- **Products variety** = distinct count of Product_Key in Fact_Sales per branch / period.
- Schema type: **Galaxy (Fact Constellation)** because three fact tables share Dim_Date and Dim_Branch. For a Snowflake variant, split Dim_Branch into Dim_Branch > Dim_Governorate > Dim_Region and Dim_Product into Dim_Product > Dim_Category.

Case Study 1 - Galaxy Schema Diagram



Case Study 2: Real Estate Agency

Step 1 - Business process

- The supervisor needs a global view of the **estates** the agency deals with and of the **agents' work**.
- Processes covered: property **visits**, **sales** and **rentals**.

Step 2 - Grain

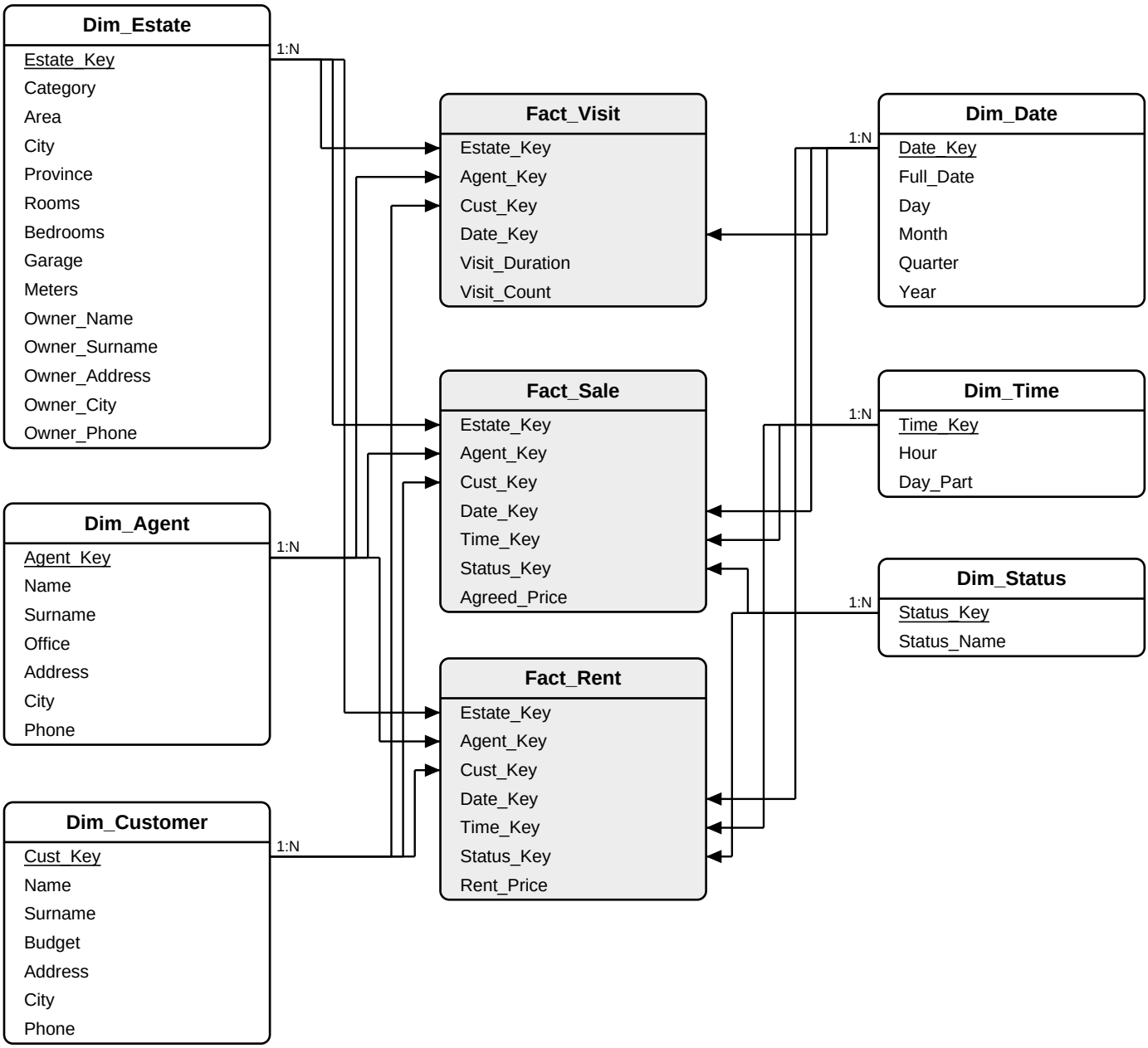
Fact table	Grain (one row =)
Fact_Visit	One visit of one customer to one estate by one agent on one date
Fact_Sale	One sale transaction (estate, agent, customer, date, time)
Fact_Rent	One rental contract (estate, agent, customer, date, time)

Step 3 - Dimensions

Dimension	Attributes
Dim_Estate	Estate_Key, Category, Area, City, Province, Rooms, Bedrooms, Garage, Meters, Owner_Name, Owner_Surname, Owner_Address, Owner_City, Owner_Phone
Dim_Agent	Agent_Key, Name, Surname, Office, Address, City, Phone
Dim_Customer	Cust_Key, Name, Surname, Budget, Address, City, Phone
Dim_Date	Date_Key, Full_Date, Day, Month, Quarter, Year
Dim_Time	Time_Key, Hour, Day_Part
Dim_Status	Status_Key, Status_Name

Step 4 - Facts (measures)

Fact table	Foreign keys	Measures
Fact_Visit	Estate, Agent, Customer, Date	Visit_Duration (additive), Visit_Count = 1
Fact_Sale	Estate, Agent, Customer, Date, Time, Status	Agreed_Price (additive), Sale_Count = 1
Fact_Rent	Estate, Agent, Customer, Date, Time, Status	Rent_Price (additive), Rent_Count = 1

Schema type: **Galaxy (Fact Constellation)** - three fact tables sharing Dim_Estate, Dim_Agent, Dim_Customer and Dim_Date.

Case Study 2 - Galaxy Schema Diagram